ENGLISH

SHARP

SCIENTIFIC CALCULATOR

WriteView

EL-W531TG
EL-W531TH
MODEL **EL-W535XG**

OPERATION MANUAL

PRINTED IN CHINA / IMPRIMÉ EN CHINE / IMPRESO EN CHINA
19ASC(TINSEA152EHM7)

INTRODUCTION

About the **calculation examples (including some formulas and tables)**, refer to the reverse side of this manual.
After reading this manual, store it in a convenient location for future reference.
Note: Some of the models described in this manual may not be available in some countries.

Operational Notes

- Do not carry the calculator around in your back pocket, as it may break when you sit down. The display is made of glass and is particularly fragile.
- Keep the calculator away from extreme heat such as on a car dashboard or near a heater, and avoid exposing it to excessively humid or dusty environments.
- Since this product is not waterproof, do not use it or store it where fluids, for example water, can splash onto it. Raindrops, water spray, juice, coffee, steam, perspiration, etc. will also cause malfunction.
- Clean with a soft, dry cloth. Do not use solvents or a wet cloth. Avoid using a rough cloth or anything else that may cause scratches.
- Do not drop it or apply excessive force.
- Never dispose of batteries in a fire.
- Keep batteries out of the reach of children.
- For the sake of your health, try not to use this product for long periods of time. If you need to use the product for an extended period, be sure to allow your eyes, hands, arms, and body adequate rest periods (about 10–15 minutes every hour).
  If you experience any pain or fatigue while using this product, discontinue use immediately. If the discomfort continues, please consult a doctor.
- This product, including accessories, may change due to upgrading without prior notice.

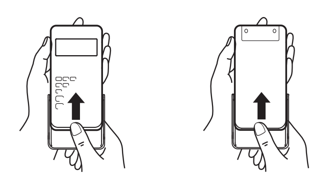
NOTICE

- SHARP strongly recommends that separate permanent written records be kept of all important data. Data may be lost or altered in virtually any electronic memory product under certain circumstances. Therefore, SHARP assumes no responsibility for data lost or otherwise rendered unusable whether as a result of improper use, repairs, defects, battery replacement, use after the specified battery life has expired, or any other cause.
- SHARP will not be liable nor responsible for any incidental or consequential economic or property damage caused by misuse and/or malfunctions of this product and its peripherals, unless such liability is acknowledged by law.

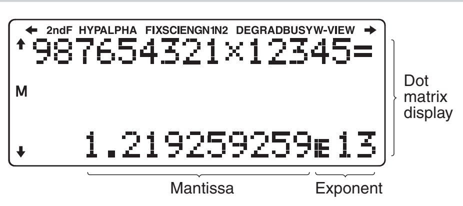
♦ Press the RESET switch (on the back), with the tip of a ball-point pen or similar object, only in the following cases. Do not use an object with a breakable or sharp tip. Note that pressing the RESET switch erases all data stored in memory.
- When using for the first time
- After replacing the battery
- To clear all memory contents
- When an abnormal condition occurs and all keys are inoperative

♦ If service should be required on this calculator, have the calculator serviced in the region (country) where you purchased it.

Hard Case



DISPLAY



- During actual use, not all symbols are displayed at the same time.
- Only the symbols required for the usage under instruction are shown in the display and calculation examples.

←/→: ↑/↓: Indicates that some contents are hidden in the directions shown.
2ndF: Appears when [2ndF] is pressed, indicating that the functions shown in the same color as [2ndF] are enabled.
HYP: Indicates that [hyp] has been pressed and the hyperbolic functions are enabled. If [2ndF] [arc hyp] is pressed, the symbols "**2ndF HYP**" appear, indicating that inverse hyperbolic functions are enabled.
ALPHA: Appears when [ALPHA] is pressed, indicating that the functions shown in the same color as [ALPHA] are enabled. Appears when [STO] or [RCL] is pressed, and entry (recall) of memory contents can be performed.
FIX/SCI/ENG/N1/N2: Indicates the notation used to display a value and changes by SET UP menu. **N1** is displayed on-screen as "NORM1", and **N2** as "NORM2".
DEG/RAD/GRAD: Indicates angular units.
BUSY: Appears during the execution of a calculation.
W-VIEW: Indicates that the WriteView editor is selected.
M: Indicates that a numerical value is stored in the independent memory (M).

BEFORE USING THE CALCULATOR

Press [ON/C] to turn the calculator on. The data that was on-screen when the power was turned off will appear on the display.
Press [2ndF] [OFF] to turn it off.

Key Notations Used in this Manual

e^x E [ln] — To specify e^x: [2ndF] [e^x]; To specify ln: [ln]; To specify E: [ALPHA] [E]

- Functions that are printed in gray adjacent to the keys are effective in specific modes.
- The multiplication operator "×" is differentiated from the letter "X" in this manual as follows:
  To specify the multiplication operator: [×]
  To specify the letter "X": [ALPHA] [X]
- In certain calculation examples, where you see the **LINE** symbol, the key operations and calculation results are shown as they would appear in the Line editor.
- In each example, press [ON/C] to clear the display first. Unless otherwise specified, calculation examples are performed in the WriteView editor ([SET UP] [2] [0] [0]) with the default display settings.

Clearing the Entry and Memories

Operation	Entry (Display)	A–F, M, X, Y	D1–D3	ANS	STAT*1
[ON/C]	O	X	X	X	X
[2ndF] [CA]	O	X	X	O	O
Mode selection ([MODE])	O	X	X	X	X*2
[2ndF] [M-CLR] [0]	O	X	X	X	X
[2ndF] [M-CLR] [1] [0]	O	O	O	O	O
[2ndF] [M-CLR] [2] [0]*3	O	O	O	O	O
RESET switch*3	O	O	O	O	O

O: Clear X: Retain

*1 Statistical data (entered data)
*2 Cleared when changing between sub-modes in STAT mode.
*3 The RESET operation will erase all data stored in memory and restore the calculator's default settings.

Memory clear key

Press [2ndF] [M-CLR] to display the menu.
- To initialize the display settings, press [0]. The parameters set as follows:
  - Angular unit: DEG
  - Display notation: NORM1
  - N-base: DEC
  - Recurring decimal: OFF

Mode Selection

NORMAL mode: [MODE] [0]
Used to perform arithmetic operations and function calculations.

STAT mode: [MODE] [1]
Used to perform statistical operations.

TABLE mode: [MODE] [2]
Used to illustrate the changes in values of a function in table format.

DRILL mode: [MODE] [3]
Used to practice math and multiplication table drills.

HOME Key

Press [HOME] to return to NORMAL mode from other modes.
Note: Equations and values currently being entered will disappear, in the same way as when the mode is changed.

SET UP Menu

Press [SET UP] to display the SET UP menu.
Press [ON/C] to exit the SET UP menu.
Note: You can press [BS] to return to the previously displayed parent menu.

Determination of the angular unit (degrees, radians, and grades)

DEG (°): [SET UP] [0] [0] (default)
RAD (rad): [SET UP] [0] [1]
GRAD (g): [SET UP] [0] [2]

Selecting the display notation and decimal places ❶

Two settings of Floating point (NORM1 and NORM2), Fixed decimal point (FIX), Scientific notation (SCI), and Engineering notation (ENG).
- When [SET UP] [1] [0] (FIX) or [SET UP] [1] [2] (ENG) is pressed, the number of decimal places (TAB) can be set to any value between 0 and 9.
- When [SET UP] [1] [1] (SCI) is pressed, the number of significant digits can be set to any value between 0 and 9. Entering 0 will set a 10-digit display.

Setting the floating point number system in scientific notation

NORM1 (the default) and NORM2. A number is automatically displayed in scientific notation outside a preset range:
NORM1 ([SET UP] [1] [3]): 0.000000001 ≤ | x | ≤ 9,999,999,999
NORM2 ([SET UP] [1] [4]): 0.01 ≤ | x | ≤ 9,999,999,999

Selecting the editor and setting the answer display ❷

This calculator has the following two editors in NORMAL mode: WriteView and Line.
Set the display format for numerical calculation results in WriteView editor.

The WriteView editor
EXACT(a/b,√,π) [SET UP] [2] [0] [0] (default)
APPROX. [SET UP] [2] [0] [1]
The Line editor [SET UP] [2] [1]

Notes:
- When "EXACT(a/b,√,π)" is set, results will appear in fraction format or irrational number format (including π and √) when display is possible.
- When "APPROX." is set, results will be decimal display or fraction display, and will be not shown in irrational number format (including π and √).
- Press [CHANGE] to change the calculation results to another format that can be displayed.

Adjusting the display contrast

Press [SET UP] [3], then [+] or [−] to adjust the contrast. Press [ON/C] to exit.

Insert and overwrite entry methods

When using the Line editor, you can change the entry method from "INSERT" (the default) to "OVERWRITE".
After you switch to the overwrite method (by pressing [SET UP] [4] [1]), the triangular cursor will change to a rectangular one, and the number or function underneath it will be overwritten as you make entries.

Setting the recurring decimal ❸

In NORMAL mode, calculation results can be shown in a recurring decimal format.
Recurring decimal is OFF: [SET UP] [5] [0] (default)
Recurring decimal is ON: [SET UP] [5] [1]
- In the WriteView editor, the recurring part is indicated by "‾". In the Line editor, the recurring part is indicated in parentheses.
- If over 10 digits, including the recurring part, the result cannot be displayed in recurring decimal format.

Setting of the decimal point

You can show the decimal point in the calculation result as either a dot or a comma.
DOT: [SET UP] [6] [0] (default)
COMMA: [SET UP] [6] [1]
- During entry, the decimal point is only shown as a dot.

ENTERING, DISPLAYING, AND EDITING THE EQUATION ❹

The WriteView Editor

Entry and display

In the WriteView editor, you can enter and display fractions or certain functions as you would write them.
- The WriteView editor can be used in NORMAL mode.

Displaying calculation results (when EXACT is selected)

When possible, calculation results will be displayed using fractions, √, and π. When you press [CHANGE], the display will cycle through the following display styles:
- Mixed fractions (with or without π) → improper fractions (with or without π) → decimal numbers
- Proper fractions (with or without π) → decimal numbers
- Irrational numbers (square roots, fractions made using square roots) → decimal numbers

Notes:
- In the following cases, calculation results may be displayed using √:
  - Arithmetic operations and memory calculations
  - Trigonometric calculations
- In trigonometric calculations, when entering values such as those in the table to the right, results may be shown using √.

	Entry value
DEG	multiples of 15
RAD	multiples of $\frac{1}{12}\pi$
GRAD	multiples of $\frac{50}{3}$

- Improper/proper fractions will be converted to and displayed as decimal numbers if the number of digits used in their expression is greater than nine. In the case of mixed fractions, the maximum number of displayable digits (including integers) is eight.
- If the number of digits in the denominator of a fractional result that uses π is greater than three, the result is converted to and displayed as a decimal number.

The Line Editor

Entry and display

In the Line editor, you can enter and display equations line by line.
Notes:
- Up to three lines of text may be viewed on the screen at one time.
- In the Line editor, calculation results are displayed in decimal form or line fraction notation if possible.
- Use [CHANGE] to switch the display format to fractional form or decimal form (if possible).

Editing the Equation

Just after obtaining an answer, pressing [◄] brings you to the end of the equation and pressing [►] brings you to the beginning. Press [◄], [►], [▲], or [▼] to move the cursor. Press [2ndF] [◄] or [2ndF] [►] to jump the cursor to the beginning or the end of the equation.

Back space and delete key

To delete a number or function, move the cursor to the right of it, then press [BS]. You can also delete a number or function that the cursor is directly over by pressing [2ndF] [DEL].
Note: In a multi-level menu, you can press [BS] to back to the previous menu level.

Multi-line Playback Function ❺

This calculator is equipped with a function to recall previous equations and answers in NORMAL mode. Pressing [▲] will display the previous equation. The number of characters that can be saved is limited. When the memory is full, stored equations will be deleted to make room, starting with the oldest.
- To edit an equation after recalling it, press [◄] or [►].
- The multi-line memory will be cleared by the following operations: [2ndF] [CA], mode change, RESET, N-base conversion, angular unit conversion, editor change ([SET UP] [2] [0] [0], [SET UP] [2] [0] [1] or [SET UP] [2] [1]), and memory clear ([2ndF] [M-CLR] [1] [0]).

Priority Levels in Calculation

This calculator performs operations according to the following priority:
① Fractions (1┏4, etc.) ② Functions preceded by their argument (x^{-1}, x^2, n!, etc.) ③ y^x, $\sqrt[x]{}$ ④ Implied multiplication of a memory value (2Y, etc.) ⑤ Functions followed by their argument (sin, cos, etc.) ⑥ Implied multiplication of a function (2sin30, A$\frac{1}{4}$, etc.) ⑦ nCr, nPr, GCD, LCM ⑧ ×, ÷, int÷ ⑨ +, − ⑩ AND ⑪ OR, XOR, XNOR ⑫ =, M+, M–, ⇒M, ► DEG, ► RAD, ► GRAD, →$r\theta$, →xy, and other calculation ending instructions
- If parentheses are used, parenthesized calculations have precedence over any other calculations.

SCIENTIFIC CALCULATIONS

- Press [MODE] [0] to select NORMAL mode.

Arithmetic Operations ❻

- The closing parenthesis [)] just before [=] or [M+] may be omitted.

Constant Calculations ❼

- In constant calculations, the addend becomes a constant. Subtraction and division are performed in the same manner. For multiplication, the multiplicand becomes a constant.
- In constant calculations, constants will be displayed as K.

Conversion to Engineering notation ❽

You can use [ALPHA] [<ENG] or [ALPHA] [ENG>] to convert the calculation result to engineering notation.
- Press [ALPHA] [<ENG] to decrease the exponent. Press [ALPHA] [ENG>] to increase the exponent.
- The settings (FSE) in the SET UP menu do not change.

Functions ❾

- Refer to the calculation examples for each function.
- In the Line editor, the following symbols are used:
  - ⌃: to indicate an expression's power. ([y^x], [2ndF] [e^x], [2ndF] [10^x])
  - ┏: to separate integers, numerators, and denominators. ([a/b], [2ndF] [ab/c])
- When using [2ndF] [$\log_a x$] or [2ndF] [abs] in the Line editor, values are entered in the following way:
  - logn (*base*, *value*)
  - abs *value*

Random Function

The random function has four settings. (This function cannot be selected while using the N-base function.) To generate further random numbers in succession, press [ENTER]. Press [ON/C] to exit.

Random numbers

A pseudo-random number, with three significant digits from 0 up to 0.999, can be generated by pressing [2ndF] [RANDOM] [0] [ENTER].
Note: In the WriteView editor, the result will be a fraction or 0.

Random dice

To simulate a die-rolling, a random integer between 1 and 6 can be generated by pressing [2ndF] [RANDOM] [1] [ENTER].

Random coin

To simulate a coin flip, 0 (heads) or 1 (tails) can be randomly generated by pressing [2ndF] [RANDOM] [2] [ENTER].

Random integer

You can specify a range for the random integer with "R.Int(" only.
R.Int(*minimum value, maximum value*)
For example, if you enter [2ndF] [RANDOM] [3] 1 [(x,y)] 99 [)] [ENTER], a random integer from 1 to 99 will be generated.

Angular Unit Conversions ❿

Each time [2ndF] [DRG►] is pressed, the angular unit changes in sequence.

Memory Calculations ⓫

Memory calculations can be performed in NORMAL and STAT modes.

Temporary memories (A–F, X and Y)

Press [STO] and a variable key to store a value in memory.
Press [RCL] and a variable key to recall the value from that memory.
To place a variable in an equation, press [ALPHA] and a variable key.

Independent memory (M)

In addition to all the features of temporary memories, a value can be added to or subtracted from an existing memory value.
Press [ON/C] [STO] [M] to clear the independent memory (M).

Last answer memory (ANS)

The calculation result obtained by pressing [=] or any other calculation ending instruction is automatically stored in the last answer memory.
Notes:
- Calculation results from the functions indicated below are automatically stored in the X or Y memories replacing any existing values.
  - →$r\theta$, →xy: X memory (r or x), Y memory (θ or y)
  - Two x' values from a quadratic regression calculation in STAT mode: X memory (1:), Y memory (2:)
- Use of [RCL] or [ALPHA] will recall the value stored in memory using up to 14 digits.

Definable memories (D1–D3)

You can store functions or operations in definable memories (D1–D3).
- To store a function or operation, press [STO], followed by a definable memory key ([D1], [D2], or [D3]), followed by the operation you want to store. Menu-related operations, such as [SET UP], cannot be stored. Press [ON/C] to return to the previous display.
- To call a stored function or operation, press the corresponding memory key. Calling a stored function will do nothing if the function that is called would be unusable in the current context.
- Any functions or operations that are stored in a definable memory will be replaced when you save a new one into that memory.
- You cannot store functions or operations in definable memories when entering values or items in STAT mode.

Memory List

Press [ALPHA] [MEMORY] to display a list of the values saved in memory.
The values are shown in a 9-character range.
Applicable memories: A, B, C, D, E, F, X, Y, M

Chain Calculations ⓬

The previous calculation result can be used in the subsequent calculation. However, it cannot be recalled after entering multiple instructions.

Fraction Calculations ⓭

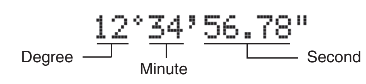
Arithmetic operations and memory calculations can be performed using fractions. In NORMAL mode, conversion between a decimal number and a fraction can be performed by pressing [CHANGE].
Notes:
- Improper/proper fractions will be converted to and displayed as decimal numbers if the number of digits used in their expression is greater than nine. In the case of mixed fractions, the maximum number of displayable digits (including integers) is eight.
- To convert a sexagesimal value to a fraction, first convert it by pressing [2ndF] [↔DEG].

Binary, Pental, Octal, Decimal, and Hexadecimal Operations (N-base) ⓮

Conversions can be performed between N-base numbers in NORMAL mode. The four basic arithmetic operations, calculations with parentheses, and memory calculations can also be performed, along with the logical operations AND, OR, NOT, NEG, XOR, and XNOR on binary, pental, octal, and hexadecimal numbers.
Note: The hexadecimal numbers A–F are entered by pressing [y^x] (A), [√] (B), [x^2] (C), [log] (D), [ln] (E), and [(x,y)] (F).
In the binary, pental, octal, and hexadecimal systems, fractional parts cannot be entered. When a decimal number having a fractional part is converted into a binary, pental, octal, or hexadecimal number, the fractional part will be truncated. Likewise, when the result of a binary, pental, octal, or hexadecimal calculation includes a fractional part, the fractional part will be truncated. In the binary, pental, octal, and hexadecimal systems, negative numbers are displayed as a complement.

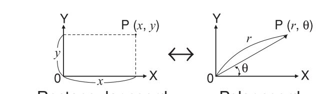
Time, Decimal, and Sexagesimal Calculations ⓯

Conversion between decimal and sexagesimal numbers can be performed. In addition, the four basic arithmetic operations and memory calculations can be performed using the sexagesimal system. Notation for sexagesimal is as follows:



Coordinate Conversions ⓰

- Before performing a calculation, select the angular unit.
- The results of coordinate conversions will be displayed as decimal numbers even in the WriteView editor.



Rectangular coord. Polar coord.

Modify Function ⓱

Decimal calculation results are internally obtained in scientific notation, with up to 14 digits in the mantissa. However, since calculation results are displayed in the form designated by the display notation and the number of decimal places indicated, the internal calculation result may differ from that shown in the display. By using the modify function ([2ndF] [MDF]), the internal value is converted to match that of the display, so that the displayed value can be used without change in subsequent operations.
- When using the WriteView editor, if the calculation result is displayed using fractions or irrational numbers, press [CHANGE] to convert it to decimal form first.

Calculating the Greatest Common Divisor (GCD)

What is the GCD of 24 and 36?	[ON/C] 24 [2ndF] [GCD] 36 [=]	12.

Calculating the Least Common Multiple (LCM)

What is the LCM of 15 and 9?	[ON/C] 15 [2ndF] [LCM] 9 [=]	45.

Calculating Quotient and Remainder ⓲

- "Q" indicates "Quotient", and "R" indicates "Remainder".
- Pressing [2ndF] [int÷] cannot be followed by pressing a key for another operation such as (+, –, ×, ÷), otherwise an error will result.
- The quotient and remainder are shown in "NORM1" format. If not all digits can be displayed in "NORM1" format, normal division is performed.

Prime Factorization ⓳

In NORMAL mode, the calculation result can be shown as a product of prime numbers.
- A positive integer greater than 2 and no more than 10 digits can be factored into primes.
- A number that cannot be factored into a prime number with 3 digits or shorter is shown in parentheses.
- The calculation result of prime factorization is displayed according to the editor setting (W-VIEW or LINE).
- The calculation result of prime factorization may extend off the edges of the screen. You can see those parts by pressing [◄] or [►]. To jump to the left end or right end, press [2ndF] [◄] or [2ndF] [►].

STATISTICAL CALCULATIONS ⓴ 21

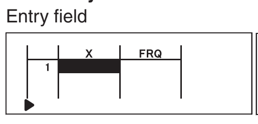
Statistical calculations can be performed in STAT mode.
There are eight sub-modes within STAT mode. Press [MODE] [1], then press the number key that corresponds to your choice:
[0] (SD): Single-variable statistics
[1] (a+bx): Linear regression
[2] (a+bx+cx²): Quadratic regression
[3] (a•e^bx): Euler exponential regression
[4] (a+b•lnx): Logarithmic regression
[5] (a•x^b): Power regression
[6] (a+b/x): Inverse regression
[7] (a•b^x): General exponential regression
The statistical data input screen appears.

After entering statistical data from the input screen, press [DATA] or [ON/C] and close the input table. You can then check statistical values from the STAT menu ([ALPHA] [STAT]) and specify statistical variables.

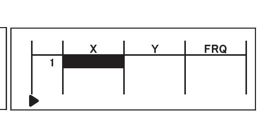
Data Entry and Correction

Data entry

Entry field



Single-variable data table



Two-variable data table

- After entering the data, press [ENTER]. The input is finalized and the cursor moves to the next line. If data was not entered in an x or y, 0 is entered, 1 is entered in FRQ (frequency), and the cursor moves to the next line.
- You can use [(x,y)] to enter X and FRQ (or X, Y, and FRQ) at once.
- In the input table, up to 6 digits are displayed for each value, including the sign and decimal point. Any values that exceed 6 digits in length are displayed in exponent notation.
- Up to 100 data items can be entered. With single-variable data, a data item with an assigned frequency of one is counted as one data item, while an item with an assigned frequency of 2 or higher is stored as a set of two data items. With two-variable data, a set of data items with an assigned frequency of one is counted as two data items, while a set of items with an assigned frequency of 2 or higher is stored as a set of three data items.
- To execute statistical calculation, press [DATA] or [ON/C] and close the input table.

Data correction

Use [◄], [►], [▲], or [▼] to move the cursor and select the desired data. Press [2ndF] [▲] or [2ndF] [▼] to jump the cursor to the beginning or end of the data.

Data correction
Move the cursor to the data that you want to correct, enter the numeric value, and press [ENTER].

Data insertion
To insert a line in front of the cursor position, press [ALPHA] [INS-D]. The initial values entered in the inserted data are 0 in x and y, and 1 in FRQ.

Data deletion
To delete the entire line where cursor is positioned, press [2ndF] [DEL].
Notes:
- In STAT mode, all statistical data will be erased if the submode is changed or [2ndF] [CA] is pressed.
- In STAT mode, press [DATA] to display the input table.

Statistical Calculations and Variables

The following statistics can be obtained for each statistical calculation (refer to the table below):

Single-variable statistical calculation

Statistics of ① and ③.

Linear regression calculation

Statistics of ①, ② and ④. In addition, the estimate of y for a given x (estimate y') and the estimate of x for a given y (estimate x').

Quadratic regression calculation

Statistics of ①, ② and ④. And coefficients a, b, c in the quadratic regression formula ($y = a + bx + cx^2$). (For quadratic regression calculations, no correlation coefficient (r) can be obtained.)
When there are two x' values, each value will be displayed with "1:" or "2:", and stored separately in the X and Y memories.
You can also specify the 1st value (x1') and the 2nd value (x2') separately.

Euler exponential regression, logarithmic regression, power regression, inverse regression, and general exponential regression calculations

Statistics of ①, ② and ④. In addition, the estimate of y for a given x and the estimate of x for a given y. (Since the calculator converts each formula into a linear regression formula before actual calculation takes place, it obtains all statistics, except coefficients a and b, from converted data rather than entered data.)

①	n	Number of samples
	$\bar{x}$	Mean of samples (x data)
	sx	Sample standard deviation (x data)
	s^2x	Sample variance (x data)
	σx	Population standard deviation (x data)
	$\sigma^2 x$	Population variance (x data)
	Σx	Sum of samples (x data)
	Σx^2	Sum of squares of samples (x data)
	$xmin$	Minimum value of samples (x data)
	$xmax$	Maximum value of samples (x data)
②	$\bar{y}$	Mean of samples (y data)
	sy	Sample standard deviation (y data)
	s^2y	Sample variance (y data)
	σy	Population standard deviation (y data)
	$\sigma^2 y$	Population variance (y data)
	Σy	Sum of samples (y data)
	Σy^2	Sum of squares of samples (y data)
	Σxy	Sum of products of samples (x, y)
	$\Sigma x^2 y$	Sum of products of samples (x^2, y)
	Σx^3	Sum of 3rd powers of samples (x data)
	Σx^4	Sum of 4th powers of samples (x data)
	$ymin$	Minimum value of samples (y data)
	$ymax$	Maximum value of samples (y data)
③	Q_1	First quartile of sample (x data)
	Med	Median of sample (x data)
	Q_3	Third quartile of sample (x data)
④	r	Correlation coefficient (Except Quadratic regression)
	a	Coefficient of regression equation
	b	Coefficient of regression equation
	c	Coefficient of quadratic regression equation
	R^2	Coefficient of determination (Quadratic regression)
	r^2	Coefficient of determination (Except Quadratic regression)

STAT Menu

After closing the input table, you can view statistical values, view regression coefficient values, and specify statistical variables from STAT menu ([ALPHA] [STAT]).
[ALPHA] [STAT] [0]: Display statistical values
[ALPHA] [STAT] [1]: Display regression coefficient values
[ALPHA] [STAT] [2]: Specify statistical value variables
[ALPHA] [STAT] [3]: Specify statistical value (Σ related) variables
[ALPHA] [STAT] [4]: Specify max/min value variables
[ALPHA] [STAT] [5]: Specify regression coefficient variables

Notes:
- List display of regression coefficient values and specification of regression coefficient variables do not appear in single-variable statistical calculation.
- Estimated values x' and y' are specified with the keys ([2ndF] [X'], [2ndF] [y']). If there are two x' values, you can specify x1' and x2' from the STAT menu ([ALPHA] [STAT] [5]) to obtain the values separately.
- In the statistical value and regression coefficient value lists, you cannot return to the menu by pressing [BS].

Statistical Calculation Formulas 22

An error will occur when:
- The absolute value of the intermediate result or calculation result is equal to or greater than 1 × 10^{100}.
- The denominator is zero.
- An attempt is made to take the square root of a negative number.
- No solution exists in the quadratic regression calculation.

TABLE MODE 23

You can see the changes in values of one or two functions using TABLE mode.

Setting a table

1. Press [MODE] [2] to enter TABLE mode.
2. Enter a function (Function1), and press [ENTER].
3. If needed, enter the 2nd function (Function2) and press [ENTER].
4. Enter a starting value (X_Start:), and press [ENTER].
   The default starting value is 0.
5. Enter a step value (X_Step:). The default step value is 1.
   - You can use [▲] and [▼] to move the cursor between the starting value and step value.
6. Press [ENTER] when you finish entering a step value. A table with a variable X and the corresponding values (ANS column) appears, displaying 3 lines below the starting value.
   If you entered two functions, the ANS1 and ANS2 columns appear. You can use [▲] and [▼] to change the X value and see its corresponding values in table format.

- The table is for display only and you cannot edit the table.
- The values are displayed up to 7 digits, including signs and a decimal point.
- Press [◄] or [►] to move the cursor to ANS column (ANS1 and ANS2 columns if you entered two functions) or X column.
- Full digits of the value on the cursor are displayed on the bottom right.

Notes:
- In a function, only "X" can be used as a variable, and other variables are all regarded as numbers (stored into the variables).
- Irrational numbers such as √ and π can also be entered into a starting value or a step value. You cannot enter 0 or a negative number as a step value.
- You can use WriteView editor when inputting a function.
- The following features are not used in TABLE mode: coordinate conversions, conversion between decimal and sexagesimal numbers, and angular unit conversions.
- It may take time to make a table, or "-------" may be displayed, depending on the function entered or conditions specified for the variable X.
- Please note that when making a table, the values for variable X are rewritten.
- Press [2ndF] [CA] or mode selection to return to the initial screen of the mode, and return to the default values for the starting value and step value.

DRILL MODE

Math Drill: [MODE] [3] [0]
Math operation questions with positive integers and 0 are displayed randomly. It is possible to select the number of questions and operator type.

Multiplication Table (× Table): [MODE] [3] [1]
Questions from each row of the multiplication table (1 to 12) are displayed serially or randomly.

To exit DRILL mode, press [MODE] and select another mode.

Using Math Drill and × Table

1. Press [MODE] [3] [0] for Math Drill or [MODE] [3] [1] for × Table.
2. **Math Drill**: Use [▲] and [▼] to select the number of questions (25, 50, or 100).
   × Table: Use [▲] and [▼] to select a row in the multiplication table (1 to 12).
3. **Math Drill**: Use [◄] and [►] to select the operator type for questions (+, −, ×, ÷, or +−×÷).
   × Table: Use [◄] and [►] to select the order type ("Serial" or "Random").
4. Press [ENTER] to start.
   When using Math Drill or × Table (random order only), questions are randomly selected and will not repeat except by chance.
5. Enter your answer. Press [ON/C] or [BS] to clear the entered number and then enter the correct answer.
6. Press [ENTER].
   - If the answer is correct, "✔" appears and the next question is displayed.
   - If the answer is wrong, "✘" appears and the same question is displayed. This will be regarded as an incorrect answer.
   - If you press [ENTER] without entering an answer, the correct answer is displayed and then the next question is displayed. This will be regarded as an incorrect answer.
7. Continue answering the series of questions by entering the answer and pressing [ENTER].
8. After you finish, press [ENTER] and the number and percentage of correct answers are displayed.
9. Press [ENTER] to return to the initial screen for your current drill.

Ranges of Math Drill Questions

The range of questions for each operator type is as follows.

+	**Addition operator**: "0 + 0" to "20 + 20"
−	**Subtraction operator**: "0 − 0" to "20 − 20"; answers are positive integers and 0.
×	**Multiplication operator**: "1 × 0" or "0 × 1" to "12 × 12"
÷	**Division operator**: "0 ÷ 1" to "144 ÷ 12"; answers are positive integers from 1 to 12 and 0, dividends of up to 144, and divisors of up to 12.
+−×÷	**Mixed operators**: Questions within all the above ranges are displayed.

ERRORS AND CALCULATION RANGES

Errors

An error will occur if an operation exceeds the calculation ranges, or if a mathematically illegal operation is attempted. When an error occurs, pressing [◄] or [►] automatically moves the cursor back to the place in the equation where the error occurred. Edit the equation or press [ON/C] to clear the equation.

Error codes and error types

ERROR 01: Syntax error
- An attempt was made to perform an invalid operation.
  Ex. 2 [+] [−] 5 [=]

ERROR 02: Calculation error
- The absolute value of an intermediate or final calculation result equals or exceeds 10^{100}.
- An attempt was made to divide by zero (or an intermediate calculation resulted in zero).
- The calculation ranges were exceeded while performing calculations.
- 0 or a negative number was entered as a step value in TABLE mode. The absolute value of a starting value or a step value equals or exceeds 10^{100} in TABLE mode.
- When the number to be factored into primes is greater than 2 and other than a 10-digit positive integer, or when the result of prime factorization is a negative number, decimal, fraction, √, or π.

ERROR 03: Nesting error
- The available number of buffers was exceeded. (There are 10 buffers for numeric values and 64 buffers for calculation instructions).

ERROR 04: Data over
- Data items exceeded 100 in STAT mode.

Alert Messages

Cannot delete!
- The selected item cannot be deleted by pressing [BS] or [2ndF] [DEL] in the WriteView editor.
  Ex. [√] 5 [►] [x^2] [◄] [BS]
  In this example, delete the exponent before attempting to delete the parentheses.

Cannot call!
- The function or operation stored in definable memory (D1 to D3) cannot be called.
  Ex. An attempt was made to recall a statistical variable from within NORMAL mode.

Buffer full!
- The equation (including any calculation ending instructions) exceeded its maximum input buffer (159 characters in the WriteView editor or 161 characters in the Line editor). An equation may not exceed its maximum input buffer.

Calculation Ranges 24

- **Within the ranges specified, this calculator is accurate to ±1 of the 10th digit of the mantissa. However, a calculation error increases in continuous calculations due to accumulation of each calculation error. (This is the same for y^x, $\sqrt[x]{}$, n!, e^x, ln, etc., where continuous calculations are performed internally.) Additionally, a calculation error will accumulate and become larger in the vicinity of inflection points and singular points of functions.**
- Calculation ranges
  ±10^{-99} to ±9.999999999 × 10^{99} and 0.
  If the absolute value of an entry or a final or intermediate result of a calculation is less than 10^{-99}, the value is considered to be 0 in calculations and in the display.

Display of results using √ (when EXACT is selected)

Calculation results may be displayed using √ when all of the following conditions are met:
- When intermediate and final calculation results are displayed in the following form:
  $\pm\frac{a\sqrt{b}}{e} \pm \frac{c\sqrt{d}}{f}$
- When each coefficient falls into the following ranges:
  1 ≤ a < 100; 1 < b < 1,000; 0 ≤ c < 100;
  1 ≤ d < 1,000; 1 ≤ e < 100; 1 ≤ f < 100
- When the number of terms in the intermediate and final calculation results is one or two.

Note: The result of two fractional terms that include √ will be reduced to a common denominator.

BATTERY REPLACEMENT

Notes on Battery Replacement

Improper handling of batteries can cause electrolyte leakage or explosion. Be sure to observe the following handling rules:
- Make sure the new battery is the correct type.
- When installing, orient the battery properly as indicated in the calculator.
- The battery is factory-installed before shipment, and may be exhausted before it reaches the service life stated in the specifications.

Notes on erasure of memory contents
When the battery is replaced, the memory contents are erased. Erasure can also occur if the calculator is defective or when it is repaired. Make a note of all important memory contents in case accidental erasure occurs.

When to Replace the Battery

EL-W531TH: If the display has poor contrast even after adjusting the display contrast, the battery requires replacement.
EL-W531TG/W535XG: If the display has poor contrast or nothing appears on the display when [ON/C] is pressed in dim lighting, even after adjusting the display contrast, it is time to replace the battery.

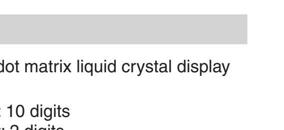
Cautions

- Fluid from a leaking battery accidentally entering an eye could result in serious injury. Should this occur, wash with clean water and immediately consult a doctor.
- Should fluid from a leaking battery come in contact with your skin or clothes, immediately wash with clean water.
- If the product is not to be used for some time, to avoid damage to the unit from a leaking battery, remove it and store in a safe place.
- Do not leave an exhausted battery inside the product.
- Keep batteries out of the reach of children.
- An exhausted battery left in the calculator may leak and damage the calculator.
- Explosion risk may be caused by incorrect handling.
- Do not throw batteries into a fire as they may explode.

Replacement Procedure

1. Turn the power off by pressing [2ndF] [OFF].
2. Remove two screws. (Fig. 1)
3. Lift the battery cover to remove.
4. **EL-W531TG/W535XG:** Remove the used battery by prying it out with a ball-point pen or other similar pointed device. (Fig. 2)
   EL-W531TH: Remove the used battery.
5. **EL-W531TG/W535XG:** Install one new battery. Make sure the "+" side is facing up.
   EL-W531TH: Install one new battery. First insert the "–" side toward the spring. (Fig. 3)
6. Replace the cover and screws.
7. Press the RESET switch (on the back) with the tip of a ball-point pen or similar object.
8. Adjust the display contrast. See "Adjusting the display contrast". And then press [ON/C].

- Make sure that the display appears as shown below. If the display does not appear as shown, remove the battery, reinstall it, and check the display once again.



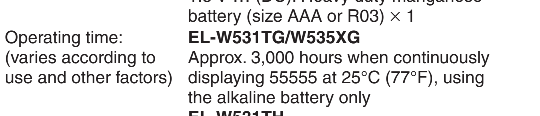


Fig. 1 Fig. 2 Fig. 3

Automatic Power Off Function

This calculator will turn itself off to save battery power if no key is pressed for approximately 10 minutes.

SPECIFICATIONS

Display:	96 × 32 dot matrix liquid crystal display
Display of calculation results:	Mantissa: 10 digits Exponent: 2 digits
Internal calculations:	Mantissas of up to 14 digits
Pending operations:	64 calculations 10 numeric values
Power source:	**EL-W531TG/W535XG** Built-in solar cells 1.5 V ⎓ (DC): Backup battery (Alkaline battery (LR44 or equivalent) × 1) **EL-W531TH** 1.5 V ⎓ (DC): Heavy duty manganese battery (size AAA or R03) × 1
Operating time: (varies according to use and other factors)	**EL-W531TG/W535XG** Approx. 3,000 hours when continuously displaying 55555 at 25°C (77°F), using the alkaline battery only **EL-W531TH** Approx. 17,000 hours when continuously displaying 55555 at 25°C (77°F)
Operating temperature:	0°C–40°C (32°F–104°F)
External dimensions:	80 mm (W) × 166 mm (D) × 15 mm (H) 3-5/32" (W) × 6-17/32" (D) × 19/32" (H)
Weight:	**EL-W531TG/W535XG** Approx. 108 g (0.24 lb) (including battery) **EL-W531TH** Approx. 113 g (0.25 lb) (including battery)
Accessories:	Battery × 1 (installed), operation manual, and hard case

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WriteView EL-W531TG EL-W531TH EL-W535XG

1 SET UP (FSE)

100000 ÷ 3 =		
[NORM1]	ON/C 100000 ÷ 3 = CHANGE CHANGE	33'333.33333
→ [FIX: TAB 2]	SET UP 1 0 2	33'333.33
→ [SCI: SIG 2]	SET UP 1 1 2	3.3E04
→ [ENG: TAB 2]	SET UP 1 2 2	33.33E03
→ [NORM1]	SET UP 1 3	33'333.33333

2 SET UP (EDITOR)

→ [APPROX.]	ON/C SET UP 2 0 1	0.
1 ÷ 2 =	1 ÷ 2 =	0.5
→ [EXACT(a/b,√,π)]	SET UP 2 0 0	0.
1 ÷ 2 =	1 ÷ 2 =	$\frac{1}{2}$

3 SET UP (RECURRING DECIMAL)

→ [ON]	ON/C SET UP 5 1	0.
611 ÷ 495 =	611 ÷ 495 =	$1\frac{116}{495}$
	CHANGE	$\frac{611}{495}$
	CHANGE	1.234
	CHANGE	1.234343434
	CHANGE	$1\frac{116}{495}$
LINE	611 ÷ 495 =	1.2(34)
	CHANGE	1.234343434
	CHANGE	1r116r495
	CHANGE	611r495
	CHANGE	1.2(34)
→ [OFF]	ON/C SET UP 5 0	0.

4 CHANGE

$\frac{2}{5}+\frac{3}{4}=$	ON/C 2 a/b 5 ▶ + a/b 3 ▶ 4 =	$1\frac{3}{20}$
	CHANGE	$\frac{23}{20}$
	CHANGE	1.15
	CHANGE	$1\frac{3}{20}$
$\sqrt{3}\times\sqrt{5}=$	√ 3 ▶ × √ 5 =	$\sqrt{15}$
	CHANGE	3.872983346
sin 45 =	sin 45 =	$\frac{\sqrt{2}}{2}$
	CHANGE	0.707106781

5 ▲ ▼

	2ndF CA	0.
① 3(5 + 2) =	3 (5 + 2) =	21.
② 3 × 5 + 2 =	3 × 5 + 2 =	17.
③ (5 + 3) × 2 =	(5 + 3) × 2 =	16.
→ ①	2ndF ▲	21.
→ ②	▼	17.
→ ①	▲	21.
→ ③	2ndF ▼	16.

6 + − × ÷ ( ) (−) Exp

45 + 285 ÷ 3 =	ON/C 45 + 285 ÷ 3 =	140.
$\frac{18+6}{15-8}=$	(18 + 6) ÷ (15 − 8 =	$3\frac{3}{7}$
42 × −5 + 120 =	42 × (−) 5 + 120 =	-90.
$(5\times10^{3})\div(4\times10^{-3})=$	5 Exp 3 ÷ 4 Exp (−) 3 =	1'250'000.

7

34 + 57 =	34 + 57 =	91.
45 + 57 =	45 =	102.
68 × 25 =	68 × 25 =	1'700.
68 × 40 =	40 =	2'720.

8 <ENG ENG>

6789=	ON/C 6789 =	6'789.
	ALPHA ENG>	6.789E03
	ALPHA ENG>	0.006789E06
	ALPHA <ENG ALPHA <ENG	6789.E00
	ALPHA <ENG	6789000.E-03

9 sin cos tan sin⁻¹ cos⁻¹ tan⁻¹ π hyp arc hyp ln log logₐx eˣ e 10ˣ x⁻¹ x² x³ yˣ √ ˣ√ ³√ n! nPr nCr % abs

sin 60 [°] =	ON/C SET UP 0 0 sin 60 =	$\frac{\sqrt{3}}{2}$
	CHANGE	0.866025403
$\cos\frac{\pi}{4}$ [rad] =	SET UP 0 1 cos π a/b 4 =	$\frac{\sqrt{2}}{2}$
	CHANGE	0.707106781
$\tan^{-1}1$ [g] =	SET UP 0 2 2ndF tan⁻¹ 1 =	50.
	SET UP 0 0	
$(\cosh 1.5+\sinh 1.5)^2=$	ON/C (hyp cos 1.5 + hyp sin 1.5) x² =	20.08553692
$\tanh^{-1}\frac{5}{7}=$	2ndF arc hyp tan (5 ÷ 7) =	0.895879734
ln 20 =	ln 20 =	2.995732274
log 50 =	log 50 =	1.698970004
$\log_2 16384$ =	2ndF logₐx 2 ▶ 16384 =	14.
LINE	2ndF logₐx 2 (x,y) 16384) =	14.
e^3 =	2ndF eˣ 3 =	20.08553692
1 ÷ e =	1 ÷ ALPHA e =	0.367879441
$10^{1.7}$ =	2ndF 10ˣ 1.7 =	50.11872336
$\frac{1}{6}+\frac{1}{7}=$	6 2ndF x⁻¹ + 7 2ndF x⁻¹ =	$\frac{13}{42}$
	CHANGE	0.309523809

$8^{-2}-3^4\times5^2=$	8 yˣ (−) 2 ▶ − 3 yˣ 4 ▶ × 5 x² =	$-2024\frac{63}{64}$
	CHANGE	$-\frac{129599}{64}$
	CHANGE	-2'024.984375
LINE	8 yˣ (−) 2 − 3 yˣ 4 × 5 x² =	-2'024.984375
	CHANGE	-2024r63r64
	CHANGE	-129599r64
8^3 =	8 2ndF x³ =	512.
$\sqrt{49}-\sqrt[4]{81}=$	√ 49 ▶ − 4 2ndF ˣ√ 81 =	4.
LINE	√ 49 − 4 2ndF ˣ√ 81 =	4.
$\sqrt[3]{27}=$	2ndF ³√ 27 =	3.
4! =	4 2ndF n! =	24.
$_{10}P_3$ =	10 2ndF nPr 3 =	720.
$_5C_2$ =	5 2ndF nCr 2 =	10.
500 × 25% =	500 × 25 2ndF %	125.
120 ÷ 400 = ?%	120 ÷ 400 2ndF %	30.
500 + (500 × 25%) =	500 + 25 2ndF %	625.
400 − (400 × 30%) =	400 − 30 2ndF %	280.
|5 − 9| =	2ndF abs 5 − 9 =	4.

	$\theta=\sin^{-1}x,\ \theta=\tan^{-1}x$	$\theta=\cos^{-1}x$
DEG	$-90\le\theta\le90$	$0\le\theta\le180$
RAD	$-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$	$0\le\theta\le\pi$
GRAD	$-100\le\theta\le100$	$0\le\theta\le200$

10 DRG▶

90° → [rad]	ON/C 90 2ndF DRG▶	$\frac{1}{2}\pi$
→ [g]	2ndF DRG▶	100.
→ [°]	2ndF DRG▶	90.

11 ALPHA RCL STO M+ M− ANS D1 D2 D3

8 × 2 ⇒ M	ON/C 8 × 2 STO M	16.
24 ÷ (8 × 2) =	24 ÷ ALPHA M =	$1\frac{1}{2}$
(8 × 2) × 5 =	ALPHA M × 5 =	80.
0 ⇒ M	ON/C STO M	0.
$150 × 3 ⇒ M_1	150 × 3 M+	450.
+) $250: M_1 + 250 ⇒ M_2	250 M+	250.
−) M_2 × 5%	RCL M × 5 2ndF % 2ndF M−	35.
M =	RCL M	665.
$\frac{24}{4+6}=2\frac{2}{5}$…(A)	24 ÷ (4 + 6) =	$2\frac{2}{5}$
3 × (A) + 60 ÷ (A) =	3 × ALPHA ANS + 60 ÷ ALPHA ANS =	$32\frac{1}{5}$
$\sinh^{-1}$ ⇒ D1	STO D1 2ndF arc hyp sin	
$\sinh^{-1}0.5$ =	D1 0.5 =	0.481211825

12

6 + 4 = ANS	ON/C 6 + 4 =	10.
ANS + 5 =	+ 5 =	15.
8 × 2 = ANS	8 × 2 =	16.
ANS² =	x² =	256.

13 a/b ab/c

$3\frac{1}{2}+\frac{4}{3}=$	ON/C 3 2ndF ab/c 1 ▼ 2 ▶ + a/b 4 ▼ 3 =	$4\frac{5}{6}$
	CHANGE	$\frac{29}{6}$
	CHANGE	4.833333333
LINE	3 a/b 1 a/b 2 + 4 a/b 3 =	4r5r6*
	CHANGE	29r6
	CHANGE	4.833333333

* 4r5r6 = $4\frac{5}{6}$

14 ▶BIN ▶PEN ▶OCT ▶HEX ▶DEC NEG NOT AND OR XOR XNOR

DEC (25) → BIN	ON/C 2ndF ▶DEC 25 2ndF ▶BIN	BIN	11001
HEX (1AC)	2ndF ▶HEX 1AC		
→ BIN	2ndF ▶BIN	BIN	110101100
→ PEN	2ndF ▶PEN	PEN	3203
→ OCT	2ndF ▶OCT	OCT	654
→ DEC	2ndF ▶DEC		428.
BIN (111) → NEG	2ndF ▶BIN NEG 111 =	BIN	1111111001
1011 AND 101 = [BIN]	2ndF ▶BIN 1011 AND 101 =	BIN	1
5A OR C3 = [HEX]	2ndF ▶HEX 5A OR C3 =	HEX	DB
NOT 10110 = [BIN]	2ndF ▶BIN NOT 10110 =	BIN	1111101001
24 XOR 4 = [OCT]	2ndF ▶OCT 24 XOR 4 =	OCT	20
B3 XNOR 2D = [HEX]	2ndF ▶HEX B3 XNOR 2D =	HEX	FFFFFFFF61
→ DEC	2ndF ▶DEC		-159.

15 D°M'S ↔DEG

7°31'49.44" → [10]	ON/C 7 D°M'S 31 D°M'S 49.44 2ndF ↔DEG	$7\frac{663}{1250}$
123.678 → [60]	123.678 2ndF ↔DEG	123°40'40.8"
3h 30m 45s + 6h 45m 36s = [60]	3 D°M'S 30 D°M'S 45 + 6 D°M'S 45 D°M'S 36 =	10°16'21."
1234°56'12" + 0°0'34.567" = [60]	1234 D°M'S 56 D°M'S 12 + 0 D°M'S 0 D°M'S 34.567 =	1234°56'47."
3h 45m − 1.69h = [60]	3 D°M'S 45 − 1.69 = 2ndF ↔DEG	2°3'36."
sin 62°12'24" = [10]	sin 62 D°M'S 12 D°M'S 24 =	0.884635235

16 →rθ →xy (x,y)

$\begin{cases}x=6\\y=4\end{cases}\rightarrow\begin{cases}r=\\\theta=[°]\end{cases}$	ON/C 6 (x,y) 4 2ndF →rθ	r: θ:	7.211102551 33.69006753
$\begin{cases}r=14\\\theta=36\ [°]\end{cases}\rightarrow\begin{cases}x=\\y=\end{cases}$	14 (x,y) 36 2ndF →xy	X: Y:	11.32623792 8.228993532

17 MDF

→ [FIX, TAB = 1]	ON/C SET UP 1 0 1	0.0
5 ÷ 9 = ANS	5 ÷ 9 =	$\frac{5}{9}$
	CHANGE	0.6
ANS × 9 =	× 9 = *1	5.0
	5 ÷ 9 =	$\frac{5}{9}$
	CHANGE	0.6
→ [MDF]	2ndF MDF	$\frac{3}{5}$
ANS × 9 =	× 9 = *2	$5\frac{2}{5}$
	CHANGE CHANGE	5.4
→ [NORM1]	SET UP 1 3	5.4

*1 $\frac{5}{9}\times9=5.5555555555555\times10^{-1}\times9$

*2 $\frac{3}{5}\times9=0.6\times9$

18 int÷

23 ÷ 5 =	ON/C 23 2ndF int÷ 5 =	Q: R:	4. 3.
9.5 ÷ 4 =	9.5 2ndF int÷ 4 =	Q: R:	2. 1.5
-32 ÷ (-5) =	(−) 32 2ndF int÷ (−) 5 =	Q: R:	6. -2.

19 P.FACT

12210 =	ON/C 12210 =	12'210.
	2ndF P.FACT	2×3×5×11×37
	2ndF P.FACT	12'210.
1234567 =	1234567 =	1'234'567.
	2ndF P.FACT	127×(9721)

20 MODE (STAT) INS-D

DATA: 20, 30, 40, 40, 50

MODE 1 0

	X	FRQ
1		

20 ENTER 30 ENTER 40 (x,y) 2 ENTER 50 ENTER

	X	FRQ
3	40	2
4	50	1
5		

DATA: 30, 40, 40, 45, 45, 45, 60

2ndF ▲ 2ndF DEL ▼ ▼ ALPHA INS-D 45 (x,y) 3 ENTER 60 ENTER

	X	FRQ
3	45	3
4	60	1
5		

21 MODE (STAT) DATA STAT x′ y′

DATA: 95, 80, 80, 75, 75, 75, 50

MODE 1 0 2ndF CA DATA 95 ENTER 80 (x,y) 2 ENTER 75 (x,y) 3 ENTER 50 ENTER

	X	FRQ
3	75	3
4	50	1
5		

DATA	Stat 0[SD]	0.
ALPHA STAT 0	n =	7.
	$\bar{x}$ =	75.7142857
	sx =	13.3630621
	s^2x =	178.571429
▼	σx =	12.3717915
	σ^2x =	153.061224
	Σx =	530.
	Σx^2 =	41'200.
▼	$xmin$ =	50.
	Q_1 =	75.
	Med =	75.
	Q_3 =	80.
▼	$xmax$ =	95.
$\frac{(95-\bar{x})}{sx}\times10+50=$	ON/C (95 − ALPHA STAT 2 1) ÷ ALPHA STAT 2 2 × 10 + 50 =	64.43210706

DATA:

x	y
2	5
2	5
12	24
21	40
21	40
21	40
15	25

MODE 1 1 2 (x,y) 5 (x,y) 2 ENTER 12 (x,y) 24 ENTER 21 (x,y) 40 (x,y) 3 ENTER 15 (x,y) 25 ENTER

	X	Y	FRQ
3	21	40	3
4	15	25	1
5			

DATA	Stat 1[a+bx]	0.
ALPHA STAT 1	$a + bx$	
	a =	1.050261097
	b =	1.826044386
	r =	0.995176343
ON/C ALPHA STAT 0 ▼ ▼ ▼ ▼ ▼	Σx^4 =	654'836.
	$ymin$ =	5.
	$ymax$ =	40.
$x=3\rightarrow y'=?$ ON/C 3 2ndF y′	3y′	6.528394256
$y=46\rightarrow x'=?$ 46 2ndF x′	46x′	24.61590706

DATA:

x	y
12	41
8	13
5	2
23	200
15	71

MODE 1 2 12 (x,y) 41 ENTER 8 (x,y) 13 ENTER 5 (x,y) 2 ENTER 23 (x,y) 200 ENTER 15 (x,y) 71 ENTER

	X	Y	FRQ
4	23	200	1
5	15	71	1
6			

DATA	Stat 2[a+bx+cx²]	0.
ALPHA STAT 1	$a + bx + cx^2$	
	a =	5.357506761
	b =	-3.120289663
	c =	0.503334057
▼	$a + bx + cx^2$, R^2 =	0.99994896
$x=10\rightarrow y'=?$ ON/C 10 2ndF y′	10y′	24.4880159
$y=22\rightarrow x'=?$ 22 2ndF x′	22x′ 1: 2:	9.63201409 -3.432772026
22 ALPHA STAT 5 5	22x′2	-3.432772026

22

$$\bar{x}=\frac{\Sigma x}{n}\qquad \sigma x=\sqrt{\frac{\Sigma x^2-n\bar{x}^2}{n}}$$

$$sx=\sqrt{\frac{\Sigma x^2-n\bar{x}^2}{n-1}}$$

$$\bar{y}=\frac{\Sigma y}{n}\qquad \sigma y=\sqrt{\frac{\Sigma y^2-n\bar{y}^2}{n}}$$

$$sy=\sqrt{\frac{\Sigma y^2-n\bar{y}^2}{n-1}}$$

23 MODE (TABLE)

x^2+1: MODE 2 ALPHA x x² + 1 ENTER ENTER

X_Start: -2, X_Step: 1: (−) 2 ENTER 1 ENTER

X	ANS
-2	5
-1	2
0	1

-2.

▼ ▼ ▼ ▼

X	ANS
0	1
1	2
2	5

2.

x^2+1: MODE 2 ALPHA x x² + 1 ENTER

$x+5$: ALPHA x + 5 ENTER

X_Start: 1, X_Step: 1: 1 ENTER 1 ENTER

X	ANS1	ANS2
1	2	6
2	5	7
3	10	8

1.

24

Function / Fonction / Funktion / Función / Funzioni / Funktion / Funktio	Dynamic range / Plage dynamique / zulässiger Bereich / Rango dinámico / Campi dinamici / Definitionsområde / Dynaaminen ala
$\sin x$, $\cos x$, $\tan x$	DEG: $\lvert x\rvert<10^{10}$ ($\tan x$: $\lvert x\rvert\ne90(2n-1)$)* RAD: $\lvert x\rvert<\frac{\pi}{180}\times10^{10}$ ($\tan x$: $\lvert x\rvert\ne\frac{\pi}{2}(2n-1)$)* GRAD: $\lvert x\rvert<\frac{10}{9}\times10^{10}$ ($\tan x$: $\lvert x\rvert\ne100(2n-1)$)*
$\sin^{-1}x$, $\cos^{-1}x$	$\lvert x\rvert\le1$
$\tan^{-1}x$, $\sqrt[3]{x}$	$\lvert x\rvert<10^{100}$
$\ln x$, $\log x$, $\log_a x$	$10^{-99}\le x<10^{100}$, $10^{-99}\le a<10^{100}$ $(a\ne1)$
y^x	• $y>0$: $-10^{100}<x\log y<100$ • $y=0$: $0<x<10^{100}$ • $y<0$: $x=n$ $(0<\lvert x\rvert<1: \frac{1}{x}=2n-1, x\ne0)$*, $-10^{100}<x\log\lvert y\rvert<100$
$\sqrt[x]{y}$	• $y>0$: $-10^{100}<\frac{1}{x}\log y<100$ $(x\ne0)$ • $y=0$: $0<x<10^{100}$ • $y<0$: $x=2n-1$ $(0<\lvert x\rvert<1: \frac{1}{x}=n, x\ne0)$*, $-10^{100}<\frac{1}{x}\log\lvert y\rvert<100$
e^x	$-10^{100}<x\le230.2585092$
10^x	$-10^{100}<x<100$
$\sinh x$, $\cosh x$, $\tanh x$	$\lvert x\rvert\le230.2585092$
$\sinh^{-1}x$	$\lvert x\rvert<10^{50}$
$\cosh^{-1}x$	$1\le x<10^{50}$
$\tanh^{-1}x$	$\lvert x\rvert<1$
x^2	$\lvert x\rvert<10^{50}$
x^3	$\lvert x\rvert<2.15443469\times10^{33}$
$\sqrt{x}$	$0\le x<10^{100}$
x^{-1}	$\lvert x\rvert<10^{100}$ $(x\ne0)$
n!	$0\le n\le69$*
$_nP_r$	$0\le r\le n\le9999999999$* $\frac{n!}{(n-r)!}<10^{100}$
$_nC_r$	$0\le r\le n\le9999999999$* $0\le r\le69$ $\frac{n!}{(n-r)!}<10^{100}$
↔DEG, D°M'S	$0°0'0.00001"\le\lvert x\rvert<10000°$
$x, y\rightarrow r, \theta$	$\sqrt{x^2+y^2}<10^{100}$
$r, \theta\rightarrow x, y$	$0\le r<10^{100}$ DEG: $\lvert\theta\rvert<10^{10}$ RAD: $\lvert\theta\rvert<\frac{\pi}{180}\times10^{10}$ GRAD: $\lvert\theta\rvert<\frac{10}{9}\times10^{10}$
DRG▶	DEG → RAD, GRAD → DEG: $\lvert x\rvert<10^{100}$ RAD → GRAD: $\lvert x\rvert<\frac{\pi}{2}\times10^{98}$
$_n$GCD$_n$, $_n$LCM$_n$	$0<n<10^{10}$*
R.Int(m, n)	$\lvert m\rvert\le9999999999$* $\lvert n\rvert\le9999999999$* $m<n$, $n-m<10^{10}$
→ DEC → BIN → PEN → OCT → HEX AND OR XOR XNOR	DEC: $\lvert x\rvert\le9999999999$ BIN: $1000000000\le x\le1111111111$, $0\le x\le111111111$ PEN: $2222222223\le x\le4444444444$, $0\le x\le2222222222$ OCT: $4000000000\le x\le7777777777$, $0\le x\le3777777777$ HEX: $FDABF41C01\le x\le FFFFFFFFFF$, $0\le x\le 2540BE3FF$
NOT	BIN: $1000000000\le x\le1111111111$, $0\le x\le111111111$ PEN: $2222222223\le x\le4444444444$, $0\le x\le2222222221$ OCT: $4000000000\le x\le7777777777$, $0\le x\le3777777777$ HEX: $FDABF41C01\le x\le FFFFFFFFFF$, $0\le x\le 2540BE3FE$
NEG	BIN: $1000000001\le x\le1111111111$, $0\le x\le111111111$ PEN: $2222222223\le x\le4444444444$, $0\le x\le2222222222$ OCT: $4000000001\le x\le7777777777$, $0\le x\le3777777777$ HEX: $FDABF41C01\le x\le FFFFFFFFFF$, $0\le x\le 2540BE3FF$

* m, n, r: integer / entier / ganze Zahlen / entero / intero / heltal / kokonaisluk

For EU only:

Manufactured by
SHARP CORPORATION
1 Takumi-cho, Sakai-ku, Sakai City,
Osaka 590-8522, Japan

Imported into Europe by
MORAVIA Europe, spol. s r.o.,
Olomoucká 83, 627 00 Brno, Czech Republic
Visit our Web site
http://www.moravia-europe.eu

Information on the Disposal of this Equipment and its Batteries

IF YOU WISH TO DISPOSE OF THIS EQUIPMENT OR ITS BATTERIES, DO NOT USE THE ORDINARY WASTE BIN ! DO NOT PUT THEM INTO A FIREPLACE !

1. In the European Union

Used electrical and electronic equipment and batteries must be collected and treated SEPARATELY in accordance with law. This ensures an environment-friendly treatment, promotes recycling of materials, and minimizes final disposal of waste. Each household should participate ! ILLEGAL DISPOSAL can be harmful to human health and the environment due to contained hazardous substances ! THIS SYMBOL appears on electrical and electronic equipment and batteries (or the packaging) to remind you of that ! If 'Hg' or 'Pb' appears below it, this means that the battery contains traces of mercury (Hg) or lead (Pb), respectively.

Take USED EQUIPMENT to a local, usually municipal, collection facility, where available. Before that, remove batteries. Take USED BATTERIES to a battery collection facility; usually a place where new batteries are sold. Ask there for a collection box for used batteries. If in doubt, contact your dealer or local authorities and ask for the correct method of disposal.

2. In other Countries outside the EU

If you wish to discard this product, please contact your local authorities and ask for the correct method of disposal.

ENGLISH

Information sur la mise au rebut de cet Équipement et de ses Piles/Batteries

SI VOUS VOULEZ METTRE AU REBUT CET ÉQUIPEMENT OU SES PILES/BATTERIES, N'UTILISEZ PAS LA POUBELLE ORDINAIRE! NE LES BRULEZ PAS DANS UNE CHEMINÉE!

1. Au sein de l'Union européenne

L'équipement électrique et électronique usagé et les piles/batteries doivent être rassemblés et traités SÉPARÉMENT conformément à la loi. Cela assure un traitement respectueux de l'environnement, promeut le recyclage de matériels et réduit au minimum le volume final de déchets. Chaque ménage devrait participer! Le DEPOT SAUVAGE peut être nuisible pour la santé humaine et l'environnement en raison de la présence de substances dangereuses! CE SYMBOLE est visible sur l'équipement électrique et électronique et sur les piles/batteries (ou sur leurs emballages) afin de vous le rappeler! Si 'Hg' ou 'Pb' apparaissent en dessous, sur la pile/batterie, cela signifie que la pile/batterie contient des traces de mercure (Hg) ou de plomb (Pb).

Déposer l'ÉQUIPEMENT USAGE à l'endroit prévu par votre municipalité, si disponible. Auparavant, ôter les Piles/Batteries usagées. Déposer des PILES/BATTERIES UTILISÉES à l'endroit prévu pour la collecte de pile/batterie; cela peut être chez votre fournisseur habituel dans le collecteur approprié. Dans le doute, entrez en contact avec votre revendeur ou les autorités locales et demandez des informations sur la méthode à utiliser pour la mise au rebut.

2. Pays hors de l'Union européenne

Si vous souhaitez mettre ce produit au rebut, veuillez contacter votre administration locale qui vous renseignera sur la méthode d'élimination correcte de cet appareil.

FRANÇAIS

Informationen zur Entsorgung dieses Gerätes und der Batterien

WENN DIESES GERÄT ODER DIE BATTERIEN ENTSORGT WERDEN SOLLEN, DÜRFEN SIE NICHT ZUM HAUSMÜLL GEGEBEN WERDEN !

1. In der Europäischen Union

Gebrauchte elektrische und elektronische Geräte und Batterien müssen laut Gesetz GETRENNT gesammelt und entsorgt werden. So werden die umweltfreundliche Abfallbehandlung und das Recycling von Stoffen sichergestellt und die Rest-Abfallmengen minimiert. Jeder Haushalt sollte dies unterstützen. Die ORDNUNGSWIDRIGE ENTSORGUNG schadet wegen der schädlichen Stoffe darin der Gesundheit und der Umwelt ! DIESES ZEICHEN auf Gerät, Batterie oder Verpackung soll Sie darauf hinweisen ! Wenn darunter noch 'Hg' oder 'Pb' steht, bedeutet dies, dass Spuren von Quecksilber (Hg) oder Blei (Pb) in der Batterie vorhanden sind.

Bringen Sie GEBRAUCHTE ELEKTRISCHE UND ELEKTRONISCHE GERÄTE zur einer örtlichen, meist kommunalen Sammelstelle, soweit vorhanden. Entnehmen Sie zuvor die Batterien. Bringen Sie VERBRAUCHTE BATTERIEN zu einer Batterie-Sammelstelle; zumeist dort, wo neue Batterien verkauft werden. Fragen Sie dort nach einem Sammelbehälter für verbrauchte Batterien. Wenden Sie sich im Zweifel an Ihren Händler oder Ihre örtlichen Behörden, um Auskunft über die richtige Entsorgung zu erhalten.

2. In anderen Ländern außerhalb der EU

Bitte erkundigen Sie sich bei Ihrer Gemeindeverwaltung nach dem ordnungsgemäßen Verfahren zur Entsorgung dieses Geräts.

DEUTSCH

Información sobre el Desechado de este Aparato y sus Pilas

SI USTED DESEA DESECHAR ESTE APARATO O SUS PILAS, NO USE EL CONTENEDOR DE RESIDUOS HABITUAL ! NO LOS DEPOSITE EN LUGARES CON FUEGO !

1. En la Unión Europea

Los aparatos eléctricos y electrónicos y las pilas usadas deben ser recogidos y tratados SEPARADAMENTE de acuerdo con la Ley. Esto garantiza un tratamiento respetuoso del medio ambiente, promueve el reciclaje de materiales, y minimiza el desecho final de residuos. Todos los hogares deben participar !. El DESECHADO ILEGAL puede ser perjudicial para la salud humana y el medio ambiente, debido a las sustancias peligrosas contenidas !. ESTE SÍMBOLO aparece en los aparatos eléctricos y electrónicos y en las pilas (o en el embalaje) para recordárselo !. Si "Hg" o "Pb" aparece debajo, significa que contienen trazas de mercurio (Hg) o plomo (Pb), respectivamente.

Lleve los APARATOS USADOS a un centro de recogida local, normalmente municipal, cuando esté disponible. Antes de eso, retire las pilas. Lleve las PILAS USADAS a un centro de recogida de pilas, por lo general un lugar donde se venden pilas nuevas. Pregunte allí por la caja de recogida de pilas usadas. En caso de duda, contacte con su distribuidor o con las autoridades locales y pregunte por el método correcto de desechado.

2. En otros países fuera de la Unión Europea

Si desea desechar este producto, por favor póngase en contacto con las autoridades locales y pregunte por el método de eliminación correcto.

ESPAÑOL

Informazioni sullo smaltimento di questo apparecchio e delle sue batterie

PER SMALTIRE IL PRESENTE DISPOSITIVO O LE SUE BATTERIE, NON UTILIZZARE IL NORMALE BIDONE DELLA SPAZZATURA ! NON INCENERIRE !

1. Nell'Unione europea

Le apparecchiature elettriche ed elettroniche usate e le batterie, devono essere raccolte SEPARATAMENTE e in conformità alla legislazione vigente Questo assicura un trattamento ambientalmente compatibile, che promuove il riciclaggio dei materiali, e minimizza il conferimento finale di rifiuti. Ognuno di noi può contribuire ! Lo SMALTIMENTO ILLEGALE può essere pericoloso per la salute umana e l'ambiente a causa delle sostanze pericolose contenute ! QUESTO SIMBOLO sulle apparecchiature elettriche ed elettroniche e le batterie o l'imballaggio è per ricordarvi questo ! Se 'Hg' o 'Pb' sono presenti là sotto significa che le batterie contengono rispettivamente tracce di mercurio (Hg) o di piombo (Pb).

Porta gli APPARECCHI USATI alla più vicina piazzola municipale ove disponibile. Rimuovi prima le batterie. Porta le BATTERIE USATE negli appositi contenitori che si trovano nelle piazzole o nei punti vendita di nuove batterie. Chiedi li per il contenitore delle batterie usate Per dubbi o chiarimenti contatta il tuo Rivenditore o le locali autorità e chiedi informazioni sul corretto metodo di smaltimento.

2. In paesi che non fanno parte dell'UE

Se si desidera eliminare il presente prodotto, contattare le autorità locali e informarsi sul metodo di smaltimento corretto.

ITALIANO

Information om avfallshantering av denna utrustning och dess batterier

OM DU ÖNSKAR AVFALLSHANTERA DENNA UTRUSTNING ELLER DESS BATTERIER, ANVÄND INTE ORDINARIE SOPTUNNA! KASTA INTE UTRUSTNINGEN ELLER DESS BATTERIER I ÖPPEN ELD!

1. EU-länder

Elektrisk och elektronisk utrustning samt batterier måste samlas in och avfallshanteras SEPARAT i enighet med gällande lagstiftning. Detta garanterar en miljövänlig hantering och återvinning av produkten och dess batterier, samt minimerar det slutliga avfallet. Varje hushåll bör hjälpa till! OLAGLIG AVFALLSHANTERING kan vara skadligt för människan och miljön på grund av farliga substanser! DENNA SYMBOL finns på elektrisk och elektronisk utrustning och batterier eller förpackning för att påminna om detta! Om texten " Hg" eller " Pb" finns nedanför, betyder detta att batteriet innehåller spår av kvicksilver (Hg) eller bly (Pb).

Ta UTRUSTNINGEN till en lokal, oftast kommunal återvinningsstation. Ta först ut batterierna. Uttjänta batterier lämnas i särskilda batteribehållare. Sådana behållare finns på alla återvinningsstationer eller i butiker där batterier säljs. Är du osäker, kontakta din återförsäljare eller kommunens miljökontor för information om korrekt avfallshantering.

2. Länder utanför EU

Kontakta de lokala myndigheterna och ta reda på gällande sorterings- och återvinningsföreskrifter om du behöver göra dig av med denna produkt.

SVENSKA

Tietoa tämän tuotteen ja sen paristojen hävittämisestä

JOS HALUAT HÄVITTÄÄ TÄMÄN TUOTTEEN TAI SEN PARISTOT, ÄLÄ HEITÄ SEKAJÄTTEIDEN JOUKKOON! EI SAA HÄVITTÄÄ POLTTAMALLA !

1. Euroopan unionissa

Käytetty elektroniikka, elektroniikkatarvikkeet ja paristot tulee kerätä ja lajitella ERIKSEEN lain määrittelemällä tavalla. Tämä takia ympäristöystävällisen jätteenkäsittelyn, parantaa materiaalien kierrätystä ja minimoi loppujätteen määrän. Jokaisen kotitalouden tulisi osallistua! LAITON HÄVITTÄMINEN voi olla haitallista terveydelle ja ympäristölle vaarallisten aineiden johdosta! TÄMÄ SYMBOLI elektroniikkatarvikkeissa ja paristoissa (tai pakkauksessa) on muistuttamassa sinua tästä! Jos alla on merkintä 'Hg' tai 'Pb', tämä tarkoittaa, että paristo sisältää jälkiä elohopeasta (Hg) tai lyijystä (Pb).

Vie KÄYTETTY ELEKTRONIIKKA paikalliseen, usein kunnan omistuksessa olevaan, jätteen käsittelylaitokseen. Ennen viemistä, poista paristot. Vie KÄYTETYT PARISTOT paristojen keräykseen; usein sijoitettu lähelle paristojen myyntipisteitä. Tiedustele keräyslaatikkoa käytetyille paristoille. Mikäli olet epävarma, ota yhteyttä myyjäliikkeeseen tai paikallisviranomaiseen ja kysy neuvoa oikeasta tavasta hävittämiselle.

2. Muissa maissa EU:n ulkopuolella

Jos haluat hävittää tuotteen, ota yhteys paikallisiin viranomaisiin ja pyydä ohjeita tuotteen asianmukaiseen hävittämiseen.

SUOMI

For Canada only :
For warranty information, please see
http://www.sharp.ca/support-product-downloads.aspx
Pour le Canada seulement :
Pour en lire plus sur la garantie, visitez le
http://www.sharp.ca/support-product-downloads.aspx

For Australia / New Zealand only :
For warranty information please see www.sharp.net.au

For Australia only:

THIS PRODUCT CONTAINS BUTTON BATTERIES. KEEP BATTERIES OUT OF REACH OF CHILDREN.